Refinement of Reliability Grid Codes in the Provision of Ancillary Services

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***Abstract*—Stochastic resources such as wind farms, electric vehicle aggregators, and demand-side assets are increasingly participating as reserve providers in ancillary service markets. To manage delivery uncertainty, system operators impose minimum reliability thresholds on such providers. Energinet, the Danish transmission system operator (TSO), has pioneered this approach through the P90 requirement, requiring stochastic providers to make accepted reserve capacity bids available with at least 90% probability. Yet this threshold is set by regulatory convention, not optimization: no existing framework treats it as a design variable or characterizes the cost-reliability trade-off it governs. This paper closes that gap. We develop a bilevel optimization framework in which the TSO in the upper level sets the reliability threshold endogenously while providers in the lower levels respond through reliability-constrained bidding, with chance constraints reformulated analytically using a Weibull tail distribution. Applied to the Nordic frequency containment reserve for disturbances (FCR-D) market, the cost-optimal threshold lies below P90 in the studied cases, with cost reductions by up to 14.5% relative to the fixed standard. Dynamic hourly thresholds yield a further reduction of up to 2.4%, suggesting efficiency gains may increase in larger and more diverse reserve markets.**

***Index Terms*—Reserve capacity procurement, stochastic energy resources, reliability grid code, chance-constrained bilevel optimization.**

NOMENCLATURE

Functions

$c_{t,i}(\cdot)$ Cost function of reserve provider i at hour t

Sets

$\mathcal{I}$ Reserve providers, $i \in I = \{1, \ldots, |\mathcal{I}|\}$

$\mathcal{T}$ Hours, $t \in \mathcal{T} = \{1, \ldots, |\mathcal{T}| = 24\}$

Ω Scenarios, $\omega \in \Omega = \{1, \ldots, |\Omega|\}$

Ω' Out-of-sample scenarios, $\omega' \in \Omega' = \{1, \ldots, |\Omega'|\}$

Parameters

$\alpha_{t,i}$,$\beta_{t,i}$ Intercept and slope coefficients of the cost function for reserve provider i at hour t [€/MW]

$1-\varepsilon^{\mathrm{sys}}$ Threshold for system-level reserve reliability [-]

$\pi_{t,\omega}$ Probability of scenario ω at hour t [-]

ρ^{sys} Penalty for reserve capacity shortfall compromising system security at hour t [€/MW]

ρ^{viol} Penalty for reserve capacity shortfall at hour t [€/MW]

D_t Minimum reserve to be procured at hour t [MW]

$R_{t,i}, r_{t,i,\omega}$ Random variable representing the reserve capacity availability of provider i at hour t, and its realization under scenario ω [MW]

Primal variables

$1-\varepsilon_t$ Threshold for reserve reliability at hour t [-]

$\hat{b}_{t,i}$ Reserve capacity bid offered by provider i at hour t [MW]

$b_{t,i}$ Cleared reserve capacity of provider i at hour t [MW]

d_t Cleared reserve demand at hour t [MW]

$s^{\mathrm{sys}}_{t,\omega}$ System-level reserve shortfall compromising system reliability under scenario ω at hour t [MW]

$s^{\mathrm{viol}}_{t,\omega}$ Total provider-level reserve shortfall under scenario ω at hour t [MW]

$s_{t,i,\omega}$ Provider i's reserve shortfall under scenario ω at hour t [MW]

$y_{t,\omega}$ Binary variable indicating system failure, where $y_{t,\omega} = 1$ if a reserve capacity shortfall occurs under scenario ω, and $y_{t,\omega} = 0$ otherwise

Dual variables

$\underline{\mu}_{t,i}, \bar{\mu}_{t,i}$ Dual variables associated with the bounds of the reserve bid by provider i at hour t [MW]

$\underline{\nu}_{t,i}, \bar{\nu}_{t,i}$ Dual variables associated with the bounds on the cleared reserve capacity of provider i at hour t [€/MW]

λ_t Reserve market-clearing price at hour t [€/MW]

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I. Mathematical model

With the set of decision variables $\boldsymbol{\xi} \in \{\varepsilon_t,\ d_t,\ s_{t,i,\omega},\ s^{\text{sys}}_{t,\omega},\ s^{\text{viol}}_{t,\omega}, y_{t,\omega}, z_{t,i}, \hat{b}_{t,i}, \bar{\mu}_{t,i}, \underline{\mu}_{t,i}, b_{t,i}, \bar{\nu}_{t,i}, \underline{\nu}_{t,i}, \lambda_t\}$, $\forall t, i, \omega$, the resulting single-level optimization problem is:

$$\min_{\boldsymbol{\varepsilon},\boldsymbol{d},\boldsymbol{s},\boldsymbol{y}} \quad \overbrace{\sum_{t,i} c_{t,i}(1-\varepsilon_t) b_{t,i}}^{\text{reserve provision cost}} + \overbrace{\sum_{t,\omega} \pi_{t,\omega}\left(\rho^{\text{viol}} s^{\text{viol}}_{t,\omega} + \rho^{\text{sys}} s^{\text{sys}}_{t,\omega}\right)}^{\text{reserve shortfall cost}} \tag{1a}$$

$$\begin{aligned}
\text{s.t.} \quad & s_{t,i,\omega} \geq 0, && \forall t,i,\omega, && (1b)\\
& s_{t,i,\omega} \geq b_{t,i} - r_{t,i,\omega}, && \forall t,i,\omega, && (1c)\\
& s^{\text{viol}}_{t,\omega} = \sum_i s_{t,i,\omega}, && \forall t,\omega, && (1d)\\
& s^{\text{sys}}_{t,\omega} \geq 0, && \forall t,\omega, && (1e)\\
& s^{\text{sys}}_{t,\omega} \geq D_t - d_t + s^{\text{viol}}_{t,\omega}, && \forall t,\omega, && (1f)\\
& d_t - s^{\text{viol}}_{t,\omega} + M y_{t,\omega} \geq D_t, && \forall t,\omega, && (1g)\\
& |\Omega| - \sum_\omega y_{t,\omega} \geq |\Omega|(1-\varepsilon^{\text{sys}}), && \forall t, && (1h)\\
& y_{t,\omega} \in \{0,1\}, && \forall t,\omega, && (1i)\\
& 0.8 \leq (1-\varepsilon_t) \leq 1, && \forall t, && (1j)\\
& \hat{b}_{t,i} \leq z_{t,i}, && \forall t,i, && (1k)\\
& -\hat{b}_{t,i} \leq 0, && \forall t,i, && (1l)\\
& -1 + \bar{\mu}_{t,i} - \underline{\mu}_{t,i} = 0, && \forall t,i, && (1m)\\
& -\bar{\mu}_{t,i} z_{t,i} \geq -\hat{b}_{t,i}, && \forall t,i, && (1n)\\
& b_{t,i} \leq \hat{b}_{t,i}, && \forall t,i, && (1o)\\
& -b_{t,i} \leq 0, && \forall t,i, && (1p)\\
& d_t - \sum_i b_{t,i} = 0, && \forall t, && (1q)\\
& \alpha_{t,i} + (1-\varepsilon_t)\beta_{t,i} - \underline{\nu}_{t,i} + \bar{\nu}_{t,i} - \lambda_t = 0, && \forall t,i, && (1r)\\
& \sum_i -\bar{\nu}_{t,i}\hat{b}_{t,i} + \lambda_t d_t \geq \sum_i \left(\alpha_{t,i} + (1-\varepsilon_t)\beta_{t,i}\right) b_{t,i}, && \forall t, && (1s)\\
& \bar{\mu}_{t,i}, \underline{\mu}_{t,i}, \bar{\nu}_{t,i}, \underline{\nu}_{t,i} \geq 0, \quad \lambda_t \text{ free}, && \forall t,i, && (1t)
\end{aligned}$$

where

$$F^{-1}\left(\frac{\varepsilon_t}{0.2}\right) =: z_{t,i}. \tag{1u}$$

Appendix

A. Reformulation of chance constraint calculations

Denoting the available reserve at the 20$^{\text{th}}$ percentile by $r^{0.2}_{t,i}$, i.e., $\mathbb{P}(R_{t,i} \leq r^{0.2}_{t,i}) = 0.2$, where, as before, $R_{t,i}$ is a continuous random variable following a distribution depending on the forecast of the individual reserve provider i at hour t, we express the conditional probability of a realization $r_{t,i,\omega}$ of the random variable $R_{t,i}$ being under the reserve providers placed bid $\hat{b}_{t,i}$ using Bayes' theorem:

$$\mathbb{P}\left(R_{t,i} \leq \hat{b}_{t,i} \mid R_{t,i} \leq r^{0.2}_{t,i}\right) = \frac{\mathbb{P}\left(R_{t,i} \leq \hat{b}_{t,i}\right)\mathbb{P}\left(R_{t,i} \leq r^{0.2}_{t,i} \mid R_{t,i} \leq \hat{b}_{t,i}\right)}{\mathbb{P}\left(R_{t,i} \leq r^{0.2}_{t,i}\right)} \tag{2a}$$

$$\Rightarrow \mathbb{P}\left(R_{t,i} \leq \hat{b}_{t,i}\right) = 0.2\mathbb{P}\left(R_{t,i} \leq \hat{b}_{t,i} \mid R_{t,i} \leq r^{0.2}_{t,i}\right), \tag{2b}$$

where it is implicit that $\mathbb{P}\left(R_{t,i} \leq r^{0.2}_{t,i} \mid R_{t,i} \leq \hat{b}_{t,i}\right) = 1$, and $\mathbb{P}(R_{t,i} \leq r^{0.2}_{t,i}) = 0.2$ as previously stated. We then have:

$$1 - \mathbb{P}\left(R_{t,i} \leq \hat{b}_{t,i}\right) \geq 1 - \varepsilon_t, \tag{2c}$$

which by (2b) and some basic arithmetic is the same as:

$$0.2\mathbb{P}\left(R_{t,i} \leq \hat{b}_{t,i} \mid R_{t,i} \leq r^{0.2}_{t,i}\right) \leq \varepsilon_t. \tag{2d}$$

B. Convexity of the CDF

For $\hat{b}_{t,i} \geq 0$ we investigate the convexity conditions of the Weibull CDF by the second derivative:

$$F(\hat{b}_{t,i}) = 1 - \exp\left(-\kappa(\hat{b}_{t,i})^\gamma\right) \tag{3a}$$

$$F'(\hat{b}_{t,i}) = f(\hat{b}_{t,i}) = \kappa\gamma(\hat{b}_{t,i})^{\gamma-1}\exp(-\kappa(\hat{b}_{t,i})^\gamma), \tag{3b}$$

$$F''(\hat{b}_{t,i}) = f'(\hat{b}_{t,i}) = \kappa\gamma(\gamma-1)(\hat{b}_{t,i})^{\gamma-2}\exp(-\kappa(\hat{b}_{t,i})^\gamma) - \kappa^2\gamma^2(\hat{b}_{t,i})^{2(\gamma-1)}\exp(-\kappa(\hat{b}_{t,i})^\gamma). \tag{3c}$$

$F(\hat{b}_{t,i})$ is strictly convex for all $\hat{b}_{t,i} \geq 0$ if $F''(\hat{b}_{t,i}) > 0$ or strictly concave if $F''(\hat{b}_{t,i}) < 0$. Canceling common terms in (3c), namely $\exp(-\kappa(\hat{b}_{t,i})^\gamma)$, $\kappa\gamma$, and $(\hat{b}_{t,i})^{\gamma-2}$, which are always non-negative, we find the convexity of $F(\hat{b}_{t,i})$ by the sign of the remaining terms:

$$\text{sign}(\gamma - 1 - \kappa\gamma(\hat{b}_{t,i})^\gamma). \tag{3d}$$

$F(\hat{b}_{t,i})$ is either convex or concave depending on the sign of $F''(\hat{b}_{t,i})$ which solely depends on γ. When $\gamma \leq 1$, $F(\hat{b}_{t,i})$ is globally concave. When $\gamma > 1$, $F(\hat{b}_{t,i})$ is convex, then concave, with the inflection point:

$$\hat{b}^*_{t,i} = \left(\frac{\gamma-1}{\kappa\gamma}\right)^{1/\gamma}, \tag{3e}$$

i.e., $F(\hat{b}_{t,i})$ is convex on $(0, \hat{b}^*_{t,i})$ and concave on $(\hat{b}^*_{t,i}, \infty)$.

C. Convexity of the inverse CDF

Let $F : (a, b) \xrightarrow{\text{onto}} (c, d) \subset \mathbb{R}$ be a convex function and let $F^{-1} : (c, d) \to \mathbb{R}$ be its inverse. Then, if F is increasing then F^{-1} is increasing and concave [1]. This intuition also applies in the other direction, from concave to convex. Due to the nature of CDFs, we know that F is an increasing function, and since the function is convex on $(0, \hat{b}^*_{t,i})$ and concave on $(\hat{b}^*_{t,i}, \infty)$, we know that its inverse is concave on $(0, \hat{b}^*_{t,i})$ and convex on $(\hat{b}^*_{t,i}, \infty)$.

D. Uniqueness of the lower-level best response

Given an isolated lower-level problem, we analyze the uniqueness of its best response given the TSO's reliability decision $1 - \varepsilon^*_t$ in the following sections.

1) Reserve providers: The optimal bid of a stochastic reserve provider is unique and given by $\hat{b}^*_{t,i} = F^{-1}\left(\varepsilon^*_t / 0.2\right)$, due to their maximization objective.

2) Market-clearing: The market-clearing problem minimizes cost subject to a demand balance constraint and capacity bounds, yielding a classical merit-order dispatch. From the stationarity condition of the Lagrangian, the optimal price must satisfy:

$$\lambda^*_t = c_{t,i}(1 - \varepsilon^*_t) + \bar{\nu}_{t,i} - \underline{\nu}_{t,i}, \quad \forall i, \tag{4a}$$

where λ^*_t is the optimal value of λ_t. Complementary slackness then partitions units into three groups: Infra-marginal units are dispatched at their upper bound $b^*_{t,i} = \hat{b}^*_{t,i}$, implying $\underline{\nu}_{i,t} = 0$ and $\bar{\nu}_{t,i} \geq 0$, from which we find:

$$\lambda^*_t \geq c_{t,i}(1 - \varepsilon^*_t). \tag{4b}$$

Extra-marginal units are shut down at $b^*_{t,i} = 0$, implying $\underline{\nu}_{i,t} \geq 0$ and $\bar{\nu}_{t,i} = 0$, from which we find:

$$\lambda^*_t \leq c_{t,i}(1 - \varepsilon^*_t). \tag{4c}$$

The marginal unit i^* clears the residual demand $b^*_{t,i^*} = d_t - \sum_{i<i^*} \hat{b}^*_{t,i}$, i.e. $0 < b^*_{t,i} < \hat{b}^*_{t,i}$, which implies $\underline{\nu}_{t,i} = \bar{\nu}_{t,i} = 0$, from which we find:

$$\lambda^*_t = c_{t,i^*}(1 - \varepsilon^*_t). \tag{4d}$$

The primal solution $b^*_{t,i}$ is unique but fails when two units share an identical cost at the margin, i.e. when $c_{t,i}(1 - \varepsilon^*_t) = c_{t,j}(1 - \varepsilon^*_t)$, $i \neq j$, producing a continuum of feasible dispatches along $b^*_{t,i} + b^*_{t,j}$, which we avoid in our case study by design. To see this via the KKTs, note that stationarity in (4a) holds with $\bar{\nu}_{t,k} = \underline{\nu}_{t,k} = 0, k \in \{i, j\}, i \neq j$ for any convex combination $(b^*_{t,i}, b^*_{t,j})$. The demand balance constraint requires only that their sum is fixed, $b^*_{t,i} + b^*_{t,j} = \text{const.}$; complementary slackness is satisfied trivially, so every point on the segment is a KKT point and hence a global optimum. Similarly, the dual variable λ^*_t is unique only when the marginal unit is strictly interior, $b^*_{t,i^*} \in (0, \hat{b}^*_{t,i^*})$; if instead demand is met exactly at a capacity bound, stationarity permits

$$\lambda^*_t \in \left[c_{t,i^*}(1 - \varepsilon^*_t),\ c_{t,i^*+1}(1 - \varepsilon^*_t)\right], \tag{4e}$$

leaving the price indeterminate within an interval. Thus, both primal dispatch and dual prices are unique, with degeneracy arising from reserve provider costs or boundary solutions, structurally possible outcomes in practice.

E. McCormick envelopes

McCormick envelopes is a type of convex relaxation that can be applied to optimization problems that contain bilinear terms of continuous variables. The relevant bilinear terms are $\varepsilon_t b_{t,i}$ in (1a), $\bar{\mu}_{t,i} z_{t,i}$ in (1n), and $\bar{\nu}_{t,i} \hat{b}_{t,i}$, $\lambda_t d_t$, and $\varepsilon_t b_{t,i}$ in (1s).

Given a bilinear expression $u_{t,i} = (1 - \varepsilon_t) b_{t,i}$ where each variable is continuous and has defined lower and upper bounds from (1j), (1o) and (1p), the new objective function with the relaxations are defined as:

$$\begin{aligned} \min_{\boldsymbol{\xi}, \boldsymbol{u}} \quad & \sum_{t,i} \alpha_{t,i} b_{t,i} + \beta_{t,i} u_{t,i} \\ & + \sum_{t,\omega} \pi_{t,\omega} \left(\rho^{\text{viol}} s^{\text{viol}}_{t,\omega} + \rho^{\text{sys}} s^{\text{sys}}_{t,\omega} \right) \end{aligned} \tag{5a}$$

with added constraints:

$$u_{t,i} \geq 0.8 b_{t,i}, \qquad \forall t, i, \tag{5b}$$

$$u_{t,i} \geq b_{t,i} - \varepsilon \hat{b}_{t,i}, \qquad \forall t, i, \tag{5c}$$

$$u_{t,i} \leq b_{t,i}, \qquad \forall t, i, \tag{5d}$$

$$u_{t,i} \leq 0.2 \hat{b}_{t,i} - \varepsilon \hat{b}_{t,i} + 0.8 b_{t,i}, \qquad \forall t, i. \tag{5e}$$

This method approximates the bilinear terms in (1) but simplifies the resulting optimization problem for computational efficiency. The method is applied to all bilinear terms in (1). For dual variables, bounds are found analytically in preparation for the relaxation (see Appendix F).

When solving for the optimal static reliability threshold, the McCormick relaxation yields a lower bound on the objective value, which compared to evaluating the original bilinear objective (1a) at the relaxed solution gives a relaxation gap of up to 4%, depending on the penalty.

Figure 1 illustrates the difference in the optimal solution using McCormick envelopes and the Pareto-optimal solution. Across the explored range of reliability values, the McCormick relaxation consistently achieves lower costs with comparable, but slightly higher, reliability levels. This gap explicitly reflects the looseness introduced by the convex relaxation. The results suggest that while the McCormick relaxation enables efficient optimization, it introduces a systematic bias that should be accounted for when interpreting solution quality.

F. Bounds on dual variables

McCormick envelopes require lower and upper bounds on the variables involved. We analytically derive bounds on dual variables that are not bounded by definition.

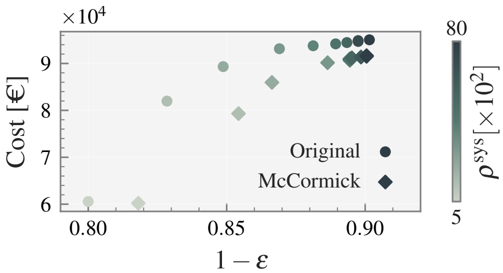

Fig. 1: Cost versus reliability for (1) (circles) and (5) (diamonds). The McCormick relaxation consistently yields lower costs at comparable reliability levels, reflecting the optimistic bias introduced by replacing bilinear terms with their convex envelope approximations.

Reserve providers: The upper bounds on the dual variables $\bar{\mu}_{t,i}$ and $\underline{\mu}_{t,i}$ are derived by exploiting the KKT stationarity condition together with the non-negativity requirements imposed by complementary slackness. Stationarity with respect to $\hat{b}_{t,i}$ yields:

$$\bar{\mu}_{t,i} - \underline{\mu}_{t,i} = 1, \tag{6a}$$

and since both multipliers must be non-negative, each is immediately bounded by:

$$0 \leq \bar{\mu}_{t,i}, \underline{\mu}_{t,i} \leq 1. \tag{6b}$$

Market-clearing: From standard economic theory, λ_t takes the value of the marginal reserve provider's cost, i.e., λ_t is bounded below by 0 and above by the cost of the most expensive reserve provider at time t:

$$0 \leq \lambda_t \leq \max_i \{c_{t,i}(1-\varepsilon_t)\}, \tag{7a}$$

which in our case is the dispatchable reserve provider with a cost independent of the reliability level. The upper bounds on $\bar{\nu}_{t,i}$ and $\underline{\nu}_{t,i}$ are derived analogously. Stationarity with respect to $b_{t,i}$ gives:

$$\bar{\nu}_{t,i} - \underline{\nu}_{t,i} = \lambda_t - c_{t,i}(1-\varepsilon_t), \quad \forall i. \tag{7b}$$

Rearranging and invoking non-negativity of both multipliers yields:

$$\bar{\nu}_{t,i} = \lambda_t - c_{t,i}(1-\varepsilon_t) + \underline{\nu}_{t,i} \geq 0, \tag{7c}$$
$$\underline{\nu}_{t,i} = c_{t,i}(1-\varepsilon_t) - \lambda_t + \bar{\nu}_{t,i} \geq 0, \tag{7d}$$

which, combined with complementary slackness enforcing that at most one multiplier per constraint pair is nonzero at optimality, gives the tight bounds:

$$0 \leq \bar{\nu}_{t,i} \leq \lambda_t - c_{t,i}(1-\varepsilon_t), \tag{7e}$$
$$0 \leq \underline{\nu}_{t,i} \leq c_{t,i}(1-\varepsilon_t) - \lambda_t. \tag{7f}$$

Substituting the upper bound on λ_t from (7a) into (7e) and (7f) yields the looser but more tractable bounds at time t:

$$0 \leq \bar{\nu}_{t,i} \leq \max_i \{c_{t,i}(1-\varepsilon_t)\}, \tag{7g}$$
$$0 \leq \underline{\nu}_{t,i} \leq \max_i \{c_{t,i}(1-\varepsilon_t)\}. \tag{7h}$$

G. Linearizations of F

As $F^{-1}(\varepsilon_t/0.2)$ is nonlinear, we linearize it for numerical simplicity. In the feasible and relevant domain of $1-\varepsilon_t \in [0.8, 1]$, we make n evenly spaced cuts and linearize between them, which Gurobi handles on its own. Hence, there is no need to implement SOS2 [2] or other methods. The definition of $F^{-1}(\varepsilon_t/0.2)$ is such that it may take negative values which is a direct violation of (1l). The linearization enables us to remove the equality condition in (1u) which is causing the violation.

References

[1] M. Mrševi, "Convexity of the inverse function," tech. rep., The Teaching of Mathematics, 2008.

[2] S. A. Gabriel *et al.*, *Complementarity Modeling in Energy Markets*. Springer New York, 2013.